B.Sc.,CS–Bachelor of Computer Science

Syllabus

Choice Based Credit System-
Learning Outcomes Based Curriculum Framework
(CBCS-LOCF)

(Applicable to the candidate admitted from the academic year 2025–2026 onwards)

B.Sc., CS- Bachelor of Computer Science Choice Based Credit System Learning Outcomes Based Curriculum Framework(CBCS-LOCF)
(Applicable to the candidate admitted from the academic year 2025–2026 onwards)

Sem.	Part	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
I	I	Language Course–I(LC) – Tamil Other Languages	Tamil-I	6	3	3	25	75	100
	II	English Language Course - I(ELC)	English-I	6	3	3	25	75	100
	III	Core Course–I(CC)	Programming in C and Data Structure	6	5	3	25	75	100
		Core Practical-I(CP)	Programming in C lab	4	4	3	50	50	100
		First Allied Course–I(AC)	Algebra & Calculus	6	4	3	25	75	100
	IV	Value Education	Value Education	2	2	3	25	75	100
	Total			**30**	**21**				**600**

Sem.	Part	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
II	I	Language Course–II(LC) - Tamil/Other Languages	Tamil-II	6	3	3	25	75	100
	II	English Language Course–II(ELC)	English-II	6	3	3	25	75	100
	III	Core Course–II(CC)	JAVA Programming	5	5	3	25	75	100
		Core Practical-II(CP)	JAVA Programming lab	4	4	3	50	50	100
		First Allied Course–II(AC)	Numerical Analysis and Probability	4	3	3	25	75	100
		First Allied Course–III(AC)	Operations Research	3	2	3	25	75	100
	IV	Environmental Studies	Environmental Studies	2	2	3	25	75	100
	Total			**30**	**22**				**700**

Sem.	PART	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
III	I	Language Course–III(LC) -Tamil/Other Languages	Tamil-III	6	3	3	25	75	100
	II	English Language Course-III (ELC)	English-III	6	3	3	25	75	100
	III	Core Course–III (CC)	Python Programming	6	5	3	25	75	100
		Core Practical-III(CP)	Python Programming lab	4	4	3	50	50	100
		Second Allied Course–I(AC)	Applied Physics–I	6	4	3	25	75	100
	IV	Non Major Elective I-	A. Fundamentals of Information Technology B. Basics of Office Automation	2	2	3	25	75	100
		ACTIVITY BASED COURSE	Health and Wellness	-	1	-	-	-	100
	Total			**30**	**22**				**700**

Sem.	Part	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
IV	I	Language Course–IV(LC) - Tamil/Other Languages	Tamil-IV	6	3	3	25	75	100
	II	English Language Course–IV (ELC)	English-IV	6	3	3	25	75	100
	III	Core Course–IV(CC)	Database Management Systems	5	5	3	25	75	100
		Core Practical-IV(CP)	Database Management Systems Lab	4	4	3	50	50	100
		Second Allied Course-II(AC)	Applied Physics–II	4	3	3	25	75	100
		Second Allied Practical–III(AP)	Applied Physics-III(Practical)	3	2	3	50	50	100
	IV	Non Major Elective II-	A. Working principles of Internet B. Fundamentals of Web Designing	2	2	3	25	75	100
	Total			**30**	**22**				**700**

Sem.	PART	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
V	III	Core Course V[CC]	Microprocessor and Microcontroller	6	5	3	25	75	100
		Core Course VI[CC]	Operating Systems	6	5	3	25	75	100
		Core Course VII[CC]	Web Technology	5	5	3	25	75	100
		Core Practical V[CP]	Microprocessor and Microcontroller Lab	4	3	3	50	50	100
		Major Based Elective-I	A. Virtualization and Cloud B. Big Data Analytics	5	4	3	25	75	100
	IV	Skill Based Elective-I	A. Amazon Web Services B. Web Designing	2	2	3	25	75	100
		Soft Skills Development	Soft Skills Development	2	2	3	25	75	100
	III	Internship	Internship i. Report of the study(80) ii. Viva voce (20)	-	2	-	20	80	100
		Total		**30**	**28**				**800**

Sem.	PART	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
VI	III	Core Course VIII[CC]	Computer Networks	6	5	3	25	75	100
		Core Course IX[CC]	Mobile Application development	6	5	3	25	75	100
		Core Practical VI[CP]	Mobile Application development Lab	5	4	3	50	50	100
		Major Based Elective-II	A. Cloud Computing B. Introduction of Data Science	5	4	3	25	75	100
		PROJECT	Project i. Project report (80) ii. Viva(20)	5	3	3	20	80	100
	IV	Skill Based Elective -II	A. Cyber Security B. Enterprise Resource Planning	2	2	3	25	75	100
	V	Extension Activities	NCC,NSS,YRC,YWF	-	1	-	-	-	-
		Gender Studies	Gender Studies	1	1	3	25	75	100
		Total		**30**	**25**				**700**
		GRAND TOTAL		**180**	**140**				**4200**

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SKILL BASED ELECTIVE–I		
S.NO	**SEMESTER**	**TITLEOFTHEPAPERS**
1.	**V**	**A.** Amazon Web Services **B.** Web Designing

SKILL BASED ELECTIVE–II		
S.NO	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**VI**	A. Cyber Security **B** . Enterprise Resource Planning

NON MAJOR ELECTIVE–I		
S.NO	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**III**	A. Fundamentals of Information Technology B. Basics of Office Automation

NON MAJOR ELECTIVE–II		
S.NO	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**IV**	**A**. Working principles of Internet **B**. Fundamentals of Web Designing

List of Allied Courses

First Allied Course I
Mathematics

Second Allied Course II
Applied Physics

SUMMARYOFCURRICULUMSTRUCTUREOFUGPROGRAMMES

S.NO	PART	TYPESOFTHECOURSE	NO.OF COURSE	NO.OF CREDITS	MARKS
1	I	Language Part–I	4	12	400
2	II	English Part–II	4	12	400
3	III	Core Paper	9	45	900
4		Core Practical	6	24	600
5		Allied Paper	5	15	500
6		Allied Practical	1	2	100
7		Major Based Elective Course	2	8	200
8		Project	1	3	100
9		Internship	1	2	100
10	IV	Non-Major Elective	2	4	200
11		Skill Based Elective	2	4	200
12		Environmental Studies	1	2	100
13		Value Education	1	2	100
14		Soft Skill Development	1	2	100
15		Activity based course	1	1	100
16	V	Gender Studies	1	1	100
17		Extension Activities	1	1	-
TOTAL			**43**	**140**	**4200**

*forthosewhostudiedTamilupto10th+2(Regular Stream)
+Syllabus for other Languages should be on par with Tamil at degree level
#thosewhostudiedTamilupto10th+2but opt for other languages in degree level under Part I should study special Tamil in Part IV

**Extension Activities shall be out side instruction hours.
Internship are 10 to 15 days.

Note:

	Internal Marks	External Marks
1.Theory	25	75
2.Practical	50	50

3. Separate passing minimum is prescribed for Internal and External marks

FOR THEORY
The passing minimum for CIA shall be 40% out of 25 marks [i.e.10marks] .
The passing minimum for University Examinations shall be 40% out of 75 marks [i.e.30marks]

FOR PRACTICAL
The passing minimum for CIA shall be 40% out of 50 marks [i.e.20 marks]
The passing minimum for University Examinations shall be 40% out of 50 marks [i.e.20marks]

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PROGRAMME OUTCOMES :

- To recognize the value of life long learning and the role of science in society.
- To identify the contemporary issues and demonstrate the scientific knowledge through various processes to arrive at solutions.
- To collect, assess and make use of the information in both theoretical and experimental science.
- To apply advanced theoretical and/or experimental methods, including the use of laboratory techniques, numerical methods and computer simulations in various fields and practical problems.
- To disseminate the science information effectively both in written or by orally before any audience.

PROGRAMME SPECIFIC OUT COMES :

- To apply the knowledge of Computer Science to identify, analyze problems and to provide effective solution in the area of Computing.
- TodevelopskillstoexcelinthefieldsofComputerScienceanditsenabledservicesinvariousfieldsof Government and Private sectors.
- To employ modern computer languages and applications for their successful career, and to create platforms to become an entrepreneur.
- To instill ethical responsibilities, human and professional values and make their contribution to the society.

FIRST YEAR	CORE COURSE–I	COURSECODE: 25CSCC1
SEMESTER I	PROGRAMMING IN C AND DATA STRUCTURE	CREDITS:5

COURSE OBJECTIVES:

- To know about the basics of C Programming.
- To know about Control and Looping Structure .
- To understand Arrays, Pointers and String Processing in C language.
- To learn about Files and its operations.
- To know about the basic concepts in Data Structures.

UNITS	CONTENTS	HOURS
I	Basic Of C : History Of C And Its Important –Structure Of C Program –Data Types-Constants And Variables-Operators And Expressions –Order Of Precedence, Evaluating Of Arithmetic Expressions-Type Conversion-Decision Statements(Various if statement and switch statement) -Managing Input And Output Statements.	20
II	Loops Structures: For Loop, While, Do-While Loop – Arrays: - One Dimensional Array, Two-Dimensional Arrays, Character Arrays And Strings – Functions: Function With Arrays-Function With Decision And Looping Statements – Recursion- Strings Processing, Standard String Library Functions .	20
III	Pointers: Introduction – Pointer Expressions – Chain Of Pointers – Pointers And Arrays – Array Of Pointers – Pointers As Function Arguments – Functions Returning Pointers – Pointers To Functions – Function Pointer – Structures - Declaration, Initialization, Array Of Structures – Pointer To Structures, Structures And Functions – Typed Of Enumerated Data Types, Unions-Files: Introduction And Files Functions – Writing And Reading In Text Mode.	20
IV	Stack: LIFO Concept, Stack Operations, Array Implementation Of Stack – Queue: FIFO Concept, Queue Operations, Array Implementation Of Queue – Singly Linked List: Concepts, Operations – Doubly Linked List: Concepts, Operations.	15
V	Trees: General Trees, Binary Trees, Binary Tree traversals-Binary Search Tree. Graph: Representation of graph-Types of Graph-Breadth first Traversal-Depth First Traversal-spanning tree -Sorting –quick sort-Insertion Sort-Merge sort-Heap sort.	15

TEXTBOOKS:

1. Szuhay, Jeff, "Learn C Programming: A Beginner's Guide to Learning C Programming the Easy and Disciplined Way",Packt Publishing, 2020.
2. C programming in Easy steps by by Mike McGrath,5th edition ,2021.
3. Data Structure with c by Seymour lipschutz ,Tata Mc Graw- hill, 6th edition 2020.
4. Fundamentals of Data Structures in C , ellishorowitz, Sartajsahni, Susan Anderson –freed , second edition , 2023 chapters-3 to 6 unit 4 & 5.

REFERENCE BOOKS:

1. E.BALAGURUSAMY, Programming in ANSI C,Tata McGraw Hill Education Private Ltd,New Delhi,Fifth Edition ,2011.
2. Data structure (concepts & implementation) ,first edition 2020 ,Technical Publications by A.A.Puntam bekar.

WEBRESOURCES:

https://www.tutorialspoint.com/cprogramming/index.html

https://www.w3schools.in/dataStructures/intro.html

https://www.prgramiz.com/c-programming

COURSEOUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
CO1	Summarize the basic knowledge to develop C programs.	K1
CO2	Manipulate Looping, arrays and functions.	K2
CO3	Apply and write programs for solving real world problems.	K3
CO4	Create open ,read ,manipulate ,write and close files.	K4
CO5	Understand the basic concepts in data structures.	K5

FIRST YEAR	CORE PRACTICAL-I	COURSECODE: 25CSCP1
SEMESTER I	PROGRAMMING IN C LAB	CREDITS:3

COURSEOBJECTIVES :

- To provide the necessary knowledge and practical training for step by step computer program development.
- To present the basic concepts in C programming language and to prepare the students to write modular and readable C Programs.
- Also the Lab Course implements the essential concepts like data types .
- To analyze and how to use such knowledge for later processing with the help of files .
- To train the students to write working programs to solve problems.

LISTOFPROGRAMS:

1. Ascending and descending order of numbers using Arrays (Use it to find Largest and Smallest Numbers).
2. Sorting of names in Alphabetical order.
3. Solution of a Quadratic Equation .
4. Sum of Series (sine, cosine).
5. Matrix operations (Addition, Subtraction, Multiplication – using functions.)
6. Finding factorials, generating Fibonacci Numbers using recursive functions.
7. String manipulations using string functions (string length, string comparison, string copy, palindrome checking, counting words and lines in strings).
8. Develop a program to create an employee details using structure and prepare a pay slip.
9. Creation and processing of Sequential files for Mark list preparation.
10. Write a program to implement the List using arrays

COURSE OUTCOMES:

CO #	CO Statement	K-Level
CO1	Understanding how problems are posed and how they can be analyzed for obtaining solutions.	K1
CO2	Understanding the fundamentals of C programming.	K2
CO3	Learning of sequencing, branching, looping and decision making statements to solve scientific and engineering problems.	K3
CO4	Implementing different operations on arrays and creating and using of functions to solve problems.	K4
CO5	Ability to design and implement different types of file structures using standard methodology.	K5

FIRST YEAR	CORE COURSE–II	COURSE CODE: 25CSCC2
SEMESTER II	JAVA PROGRAMMING	CREDITS: 5

COURSE OBJECTIVES:

- To learn how to implement object-oriented designs with Java.
- To identify Java language components and how they work together in applications.
- To design and program stand-alone Java applications.
- To learn how to use exception handling in Java applications.
- To learn about AWT concepts and Applets.

UNITS	CONTENTS	HOURS
I	Java Language Fundamentals : The Building Blocks of Java – Data Types – Variable Declarations – Wrapper Classes – Operations and Assignment – Control Structures – Arrays and its types.	15
II	Java as an OOP Language : Defining Classes – objects - Modifiers –Constructors- Inheritance and its types- Polymorphism - Packages – Interfaces - Strings class– concatenation-comparison-substring -String Buffer Class.	15
III	Exception Handling : Introduction – Basics of Exception Handling – Exception Hierarchy – Constructors and Methods in Throw able Class - Unchecked and Checked Exceptions – Handling Exceptions in Java – Exception and Inheritance – Advantages of Exception Handling Mechanism.	15
IV	Multithreading : Introduction – Creating Threads – Thread Life-cycle .Files and I/O Streams : Overview – Java I/O – File Streams – File Input Stream and File Output Stream – File Streams – Random Access File – Serialization.	15
V	Applets: Basis - Lifecycle - Applet classes - Application – Graphics; AWT-I: GUI Programming - AWT controls - Windows fundamentals- Creating Windows - Dialog Boxes - Layout Managers - Radio Buttons and Check Boxes – Borders –Java Swing.	15

TEXTBOOKS:

1. Object Oriented Programming through Java, P. Radha Krishna, University Press, 2019.

2. S.Sagayaraj, R.Denis, P.Karthik&D.Gajalakshmi, "Java Programming", Universities Press, 2021.

3. PatrickNaughton and Herbert Schildt. "The Complete Reference JAVA 2".3 rd Edition.Tata McGraw-Hill Edition, 2011.

4. Muthu C. "Programming with JAVA",2nd Edition. Vijay Nicole Imprints, 2020.

REFERENCE BOOKS:

1. Java Programming, K. Rajkumar, Pearson India, 2013
2. Muthu C. "Programming with JAVA". 2nd Edition. Vijay Nicole Imprints, 2011.
3. Ken Arnold Gosling and Davis Holmen. "The Java Programming Language". 3rd Edition. Addition Wesley Publication.

WEB RESOURCES:

https://www.tutorialspoint.com/cprogramming/index.html

https://www.w3schools.in/dataStructures/intro.html

https://www.programiz.com/java-programming

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
C01	Remember the fundamentals of programming such as variables, Conditional statements and iterative execution statements.	K1
C02	Understanding the OOPs concepts and String	K2
C03	Apply the concepts of Exception Handling.	K3
C04	Implement the concepts of Multithreading and File Streams.	K4
C05	Analyze the concepts of applet programming, graphics programming, Java swing..	K5

FIRST YEAR	CORE PRACTICAL- II	COURSE CODE: 25CSCP2
SEMESTER II	JAVA PROGRAMMING LAB	CREDITS:3

COURSE OBJECTIVES:

- To Impart Practical Training in Java Programming Language.
- To gain practical knowledge in Exception Handling.
- To gain practical expertise in coding Core Java programs.
- To become proficient in the use of AWT.
- To develop a program based on Event Handling and Swing.

LIST OF PROGRAMS:

1.Write a program to sort the given numbers using arrays.

2.Write a program to implement the FIND and REPLACE operations in the given multiple text.

3.Write a program to perform basic arithmetic Operations.

4.Writeaprogram to find the area of a rectangle using constructor.

5.Write a program to find the student's percentage and grade using command line arguments.

6.Write a program to draw circle or triangle or square using polymorphism and inheritance.

7.Implementing User defined Exception Handling.

8.Implement multiple inheritance concepts in java using interface, you can choose your own example of a company or education institution or a general concept which requires the use of interface to solve a particular problems.

9.Writeaprogramtocreatethreadsandassignprioritiestothem.

10.Write a program to develop an applet to play multiple audio clips using multithreading.

11.Write a program to implementing a Calculator using Java Swing.

COURSE OUTCOMES:

CO #	CO Statement	K-Level
CO1	Remember the fundamentals of Java programming language	K1
CO2	Understand the basics of Java programming, multi-threaded programs and Exception handling	K2
CO3	Analyze and use Java in a variety of applications.	K3
CO4	Write and debug a software application developed using the Java Programming language.	K4
CO5	Ability to design application programs using Java	K5

SECOND YEAR	CORE COURSE–III	COURSE CODE: 25CSCC3
SEMESTER III	PYTHON PROGRAMMING	CREDITS: 5

COURSE OBJECTIVES:

- To make students understand the concepts of Python programming
- To provide solutions using control structures in Python programming
- To apply the knowledge functions, strings and modules in Python based solutions
- To learn the various element-based data types in Python programming
- To work with file-based operations with Python

UNITS	CONTENTS	HOURS
I	**Basics of Python Programming:** History of Python-Features of Python-Literal-Constants-Variables - Identifiers - Keywords-Built-in Data Types - Output Statements - Input Statements - Comments - Indentation - Operators-Expressions-Type conversions. **Python Arrays:** Defining and Processing Arrays - Array methods.	15
II	**Control Statements:** Selection/Conditional Branching statements: if, if-else, nested if and if-elif-else statements. **Iterative Statements**: While loop, For loop, Else suite in loop and Nested loops. **Jump Statements:** Break, Continue and Pass statements**.**	15
III	**Functions in Python** **Functions:** Function Definition - Function Call - Variable Scope and its Lifetime-Return Statement. **Function Arguments**: Required Arguments, Keyword Arguments, Default Arguments and Variable Length Arguments - Recursion. **Python Strings:** String operations-Immutable Strings - Built-in String Methods and Functions - String Comparison. **Modules**: import statement- The Python module - dir() function - Modules and Namespace - Defining our own modules.	15
IV	**Lists and Dictionaries :Lists:** Creating a list -Access values in List-Updating values in Lists-Nested lists -Basic list operations-List Methods. Tuples: Creating, Accessing, Updating and Deleting Elements in a tuple - Nested tuples- Difference between lists and tuples. **Dictionaries:** Creating, Accessing, Updating and Deleting Elements in a Dictionary - Dictionary Functions and Methods - Difference between Lists and Dictionaries.	15
V	**File Handling :**Types of files in Python - Opening and Closing files-Reading and Writing files: write() and writelines() methods- append() method - read() and readlines() methods - with keyword - Splitting words - File methods - File Positions- Renaming and deleting files.	15

TEXTBOOKS:

1. Thareja, R. (2020). Python programming using problem solving approach, (1st Ed.).Oxford University Press.
2. Rao, N. R. (2020). Core Python programming, (1st Ed.). Dream tech Publishers.

REFERENCE BOOKS:

1. Kurama, V. (2018). Python programming: A modern approach. Pearson Education.
2. Lambert, K. A. (2017). Fundamentals of Python - First programs. CENGAGE
3. Publication.

WEB RESOURCES:

1. https://www.programiz.com/python-programming
2. https://www.guru99.com/python-tutorials.html
3. https://www.w3schools.com/python/python_intro.asp
4. https://www.geeksforgeeks.org/python-programming-language/
5. https://en.wikipedia.org/wiki/Python_(programming_language)

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
CO1	Recall simple Python programs that solve basic problems	**K1**
CO2	Explain the basic concepts of Python programming	**K2**
CO3	Use Python to interact with the operating system and other external resources.	**K3**
CO4	Analyze and apply solutions to problems by using various Python techniques.	**K4**
CO5	Develop reusable and maintainable Python software.	**K5**

FIRST YEAR	CORE PRACTICAL- III	COURSE CODE: 25CSCP3
SEMESTER III	PYTHON PROGRAMMING LAB	CREDITS:3

COURSE OBJECTIVES:

- Acquire programming skills in core Python.
- Acquire Object-oriented programming skills in Python.
- Develop the skill of designing graphical-user interfaces(GUI)in Python.
- Develop the ability to write database applications in Python.
- Acquire Python programming skills to move in to specific branches

LIST OF PROGRAMS:

1. Write a Python Program to Print a Multiplication Table for the given number.
2. Write a Python Program to check whether the given number is prime or not.
3. Write a Python Program to display the Fibonacci series for a given number.
4. Write a Python Program to implement List Operations (Nested List, Length, Concatenation, Membership, Iteration, Indexing and slicing).
5. Write a Python Program to Transpose the Matrix.
6. Write a Python Program to implement the simple Calculator.
7. Write a Python Program to Generate the Calendar for the Given Month and Year.
8. Write a Python Program to implement the List Methods (Add, Append, Extend &Delete)
9. Write a Python Program to implement the Breadth first Search Traversal.
10. Write a Python Program to sort the sentence in the alphabetical order.
11. Write a Python Program to implement the Water jug Problem.
12. Write a Python Program to Find factorial of the given number to implement the simple chat bot.

COURSE OUTCOMES:

CO #	CO Statement	K-Level
CO1	To understand the problem solving approaches	K1
CO2	To learn the basic programming on struts in Python	K2
CO3	To practice various computing strategies for Python-based solution store al world problems	K3
CO4	To use Python data structures -lists, tuples, dictionaries.	K4
CO5	To do input/output with files in Python.	K5

SECOND YEAR	NON MAJOR ELECTIVE- I	COURSE CODE: 25CSNM1A
SEMESTER III	A. FUNDAMENTALS OF INFORMATION TECHNOLOGY	CREDITS: 2

COURSE OBJECTIVES:

- To familiarize the students with the world of IT and IT-enabled services.
- To provide a in-depth knowledge about internet and internet tools.
- To enable the students to understand about Computer Security

UNITS	CONTENTS	HOURS
I	Introduction to Computers-Generation of Computers-Classification of Digital Computer - Anatomy of Digital Computer.	6
II	CPU and Memory-Secondary Story Devices-Input Devices-Output Devices.	6
III	Introduction to Computer Software-Programming Language-Operating Systems -Introduction to Database Management System.	6
IV	Computer Networks-WWW and Internet-Email-Web Design.	6
V	Computers at Home, Education, Entertainment, Science, Medicine and Engineering - Introduction to Computer Security - Computer Viruses, Bombs, Worms.	6

TEXTBOOKS:

1. **Alexis Leon & Mathews Leon** (2019), *Fundamentals of Information Technology*, Leon Tech World, Chennai.

REFERENCE BOOKS:

1. **V. Rajaraman** (2018), *Introduction to Information Technology*, PHI Learning Pvt. Ltd., New Delhi.
2. **P.K. Sinha & Priti Sinha** (2017), *Computer Fundamentals*, BPB Publications, New Delhi.
3. **V. Rajaraman & Neeharika Adabala** (2015), *Fundamentals of Computers*, PHI Learning Pvt. Ltd., New Delhi.

WEB RESOURCES:

1. Computer Fundamentals Tutorial – TutorialsPoint
2. Information Technology and Computer Basics – GeeksforGeeks
3. Computer Science and IT Resources – IBM SkillsBuild

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
CO1	Understand basic concepts and terminologies in IT and IT-enabled services.	**K1**
CO2	Understanding personal computers and their operations.	**K2**
CO3	Understand about operating systems and database management	**K3**
CO4	Comprehend about WWW internet, email and web design concepts	**K4**
CO5	Respond to computer security issues.	**K5**

SECOND YEAR	NON MAJOR ELECTIVE I	COURSE CODE: 25CSNM1B
SEMESTER III	B.BASICS OF OFFICE AUTOMATION	CREDITS: 2

COURSE OBJECTIVES:

- To create the professional-quality documents.
- To create dynamic slide presentations with animation digitally and effectively.
- To expose the students with organize and analyze information in worksheet.

UNITS	CONTENTS	HOURS
I	Introduction to Microsoft Office Navigate the Microsoft Office command structure using the Ribbon- Quick Access Toolbar Minimum Toolbar and Shortcut Menus - Use Microsoft Office help.	6
II	Ms. Word: Format text- paragraphs and document elements-Identify and use correct sequencing and spacing of all required business letter components- Format a research paper with footnotes and bibliographical list of sources-Insert a Word table, enter data and format the table- Streamline document creation using templates Insert and format a graphic or word art in Word documents.	6
III	Ms. Excel: Enter text-numbers and formulas into a worksheet applying to business situations- Create and edit Excel charts to graphically present numerical data- Format a worksheet using number formats, borders, color, fonts and conditional formats- Use the Excel SUM, AVERAGE, MAX, MIN, IF, and NOW functions to display desired calculations- Create and copy formulas using relative and absolute cell references- Edit worksheets and add new worksheets to a workbook, rename and rearrange sheets.	6
IV	Ms. Access Create a database file consisting of at least two tables using proper designing principles- Create a report that presents the data in the database in an organized and flexible manner- Sort the records of a database and use criteria to search for specific information- Create queries to retrieve specific information from one or more related tables.	6
V	Ms. PowerPoint Create single and multi-level bulleted list slides- Use the PowerPoint ribbon to change Layouts and Themes in presentations- Insert and size clip art, photographs, and shapes into a presentation- Create and run a presentation with slide transitions	6

TEXTBOOKS:

1. Microsoft Office 2016: In Practice 1st Edition by Randy Nordell.

2. Exploring Microsoft Office: The Illustrated, Practical Guide to Using Office and Microsoft by Kevin

Wilson

REFERENCE BOOKS:

1. **Shelly Cashman Series** (2020), *Microsoft Office 365 & Office 2019 Introductory*, Cengage Learning.

1. **Patricia Wallace & Ruth Maran** (2018), *Microsoft Office 2019 Simplified*, Wiley Publications.
2. **Jennifer Duffy** (2019), *Microsoft Office 2019 In Depth*, Que Publishing.
3. **Stephen L. Nelson** (2019), *Office 2019 For Dummies*, Wiley Publishing.

WEB RESOURCES:

1. Microsoft Office Support Center
2. Microsoft Learn – Office Training
3. GCFGlobal Microsoft Office Tutorials

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
CO1	Recall the features, tools, and command structure of Microsoft Office applications.	**K1**
CO2	Explain and demonstrate the use of MS Word for creating, formatting, and editing professional documents.	**K2**
CO3	Apply MS Excel functions, formulas, charts, and worksheet management techniques to solve business-related problems.	**K3**
CO4	Analyze and organize data using MS Access by creating databases, tables, queries, and reports.	**K4**
CO5	Evaluate and design effective presentations using MS PowerPoint with appropriate layouts, themes, graphics, and slide transitions.	**K5**

SECOND YEAR	CORE COURSE–IV	COURSE CODE: 25CSCC4
SEMESTER IV	DATABASE MANAGEMENT SYSTEM	CREDITS: 5

COURSE OBJECTIVES:

- To enable the students to learn the designing of database systems, foundation on the relational model of data and normal forms.
- T understood the concepts of database management system, design simple Database models
- To learn and understand to write queries using SQL,PL/SQL.

UNITS	CONTENTS	HOURS
I	Database Basics: Introduction: Flat File – Database System – Database – Actionable for DBA. The Entity – Relationship Model: Introduction – The Entity Relationship Model. Data Models: Introduction – Relational Approach – The Hierarchical Approach – The Network Approach.	15
II	Relational Algebra: Structure of Relational Databases – Fundamental Relational Algebra Operations –Additional Relational Algebra Operations - Extended Relational Algebra Operations- Null Values - Modification of the Database - The Tuple Relational Calculus – The Domain Relational Calculus.	15
III	Normalization: Normalization: Introduction - Normalization – Definition of Functional Dependence(FD) – Normal Forms: 1NF, 2NF, 3NF and BCNF.	15
IV	Structured Query Language: Structured Query Language: Features of SQL – Select SQL Operations – Grouping the Output of the Query – Querying from Multiple Tables – Retrieval Using Set operators – Nested Queries. T-SQL – Triggers and Dynamic Execution: Transact- SQL.	15
V	Procedural Language- SQL: PL/SQL Block Structure – PL/SQL Tables. Cursor Management and Advanced PL/SQL: Opening and Closing a Cursor – Processing Explicit Cursor – Implicit Cursor – Exception Handlers – Sub Programs in – Functions – Precaution While Using PL/SQL Functions – Stored Procedure – Object Oriented Technology.	15

TEXTBOOKS:

1. Coronel, Morris, Rob, "Database Systems, Design, Implementation and Management", Ninth Edition,2022
2. NileshShah, "DatabaseSystemsUsingOracle",2ndedition,PearsonEducationIndia, 2020.

REFERENCE BOOKS:

1. AbrahamSilberschatz,HenryF.KorthandS.Sudarshan,"DatabaseSystem Concepts", McGraw Hill International Publication ,VI Edition.
2. ShioKumarSingh, "Database Systems",Pearsonpublications,IIEdition

WEB RESOURCES:

Web resources from NDL Library, E-content from open-source libraries

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
C01	Write queries to manipulate data.	K1
C02	Demonstrate the aggregate functions and set operations.	K3
C03	Apply the join operations.	K3
C04	Know about usage of nested sub queries.	K4
C05	Understand the method to create views.	K5

SECOND YEAR	CORE PRACTICAL- IV	COURSE CODE: 25CSCP4
SEMESTER IV	DATABASE MANAGEMENT SYSTEM LAB	CREDITS:3

COURSE OBJECTIVES:

- To understand the basic concepts and the applications of database systems using MYSQL.
- To create and perform basic operation with MYSQL.
- To interact with MYSQL by using nested queries, set of aggregate operations

and views

LIST OF PROGRAMS:

1. Create a table and perform the following basic mysql operations
   a. Set the primary key
   b. Alter the structure of the table
   c. Insert values
   d. Delete values based on constraints
   e. Display values using various forms of select clause
   f. Drop the table
2. Develop my sql queries to implement the following set operations
   a. Union
   b. Union all
   c. Intersect
   d. Intersect all
3. Develop my sql queries to implement the following aggregate functions
   a. Sum
   b. Count
   c. Average
   d. Maximum
   e. Minimum
   f. Group by clause & having clause

I. PL/SQL

4. FIBONACCISERIES
5. FACTORIAL
6. STRING REVERSE
7. SUMOFSERIES
8. TRIGGER

II. CURSOR

9. STUDENT MARK ANALYSIS USING CURSOR

III. APPLICATION

10. LIBRARY MANAGEMENT SYSTEM

COURSE OUTCOMES:

CO #	CO Statement	K-Level
CO1	Write queries to manipulate data.	K1
CO2	Demonstrate the aggregate functions and set operations.	K2
CO3	Apply the join operations.	K3
CO4	Know about usage of nested sub queries.	K4
CO5	Understand the method to create views.	K5

SECOND YEAR	NON MAJOR ELECTIVE II	COURSE CODE: 25CSNM2A
SEMESTER IV	A.WORKING PRINCIPLES OF INTERNET	CREDITS: 2

COURSE OBJECTIVES:

- To teach the basics of the World Wide Web
- To Understand the fundamentals of the Internet and the usage
- To know the components of Multimedia on the internet

UNITS	CONTENTS	HOURS
I	What is Internet? The Internet's underlying Architecture	6
II	Connecting to the Internet– Communicating on the Internet	6
III	How the World Wide Web works .Common Internet tools	6
IV	Multimedia on the Internet–Intranet and shopping on the Internet	6
V	Safeguarding the Internet	6

TEXTBOOKS:

- Internet & World Wide Web: How to Program **Authors:** Harvey Deitel and Paul Deitel **Publisher:** Pearson Education Covers Internet fundamentals, web technologies, web communication, and Internet applications.

- The Internet Book: **Author:** Douglas Comer Provides a clear introduction to Internet architecture, protocols, and networking concepts.

REFERENCE BOOKS:

1. PrestonGralla,"HowtheInternetWorks",PearsonEducation,EighthEdition,2006.
2. C.Xavier, Fundamentals of Internet and Emerging Technologies, New Age International
3. AlexisLeon,InternetforEveryone,S.Chand(G/L)&CompanyLtd;SecondEdition2012.
4. AndreaC.Nakaya,"InternetandSocialMediaAddiction",ReferencePointPress,2015.

WEB RESOURCES:

- Internet Society (Internet Basics)
- World Wide Web Consortium (W3C)
- Mozilla Developer Network (MDN Web Docs)

COURSE OUTCOMES:

CO #	CO Statement	K-Level
CO1	Explain Internet concepts and architecture	**K1**
CO2	Describe Internet connectivity and communication	**K2**
CO3	Explain WWW and Internet tools	**K3**
CO4	Understand multimedia, intranet, and online shopping	**K4**
CO5	Apply Internet safety and security practices	**K5**

SECOND YEAR	NON MAJOR ELECTIVE- II	COURSE CODE: 25CSNM2B
SEMESTER IV	B. FUNDAMENTALS OF WEB DESIGNING	CREDITS: 2

COURSE OBJECTIVES:

- To understand the basic concepts of computer networks, client–server architecture, and web technologies.
- To learn Internet fundamentals including IP addressing, domain names, and communication protocols (HTTP, FTP, SMTP, etc.).
- To gain knowledge of web browsing, search engines, email systems, and web navigation techniques.
- To design web pages using HTML including text formatting, lists, images, links, and tables.
- To understand XML fundamentals and differentiate between HTML .

UNITS	CONTENTS	HOURS
I	Introduction to Web related terms :Client –Server–Single Tier Architecture–Two Tier Architecture– N Tier Architecture – Internet – Intranet – WWW – Web Page – Web Site.	6
II	Introduction to Internet and its Protocols: Internet Addressing – IP Address – Domain Name – Electronic Mail–Uniform Resource Locator –TCP/IP-FTP–HTTP–SHTTP-SMTP-POP–UDP – PPP – Telnet – Gopher – WAIS.	6
III	SearchingtheWeb:Browser:Web Index–WebSearch Engine–Meta Search Sites–Directoryand Indexes – Managing Mails: Address Book – File Attachment Facility – Junk Mail.	6
IV	Introduction to HTML: HTML Code for Web Page–HTML Tags-Web Page Basics–Display a Web Page in a Web Browser – Format texts – Font Style and sizes – Lists – Ordered Lists – Unordered Lists – Image – Add an image – Background Image – Horizontal Rule	6
V	Introduction to HTML Table: <TABLE> - <TH> - <TD> - <TR> - Cell Padding – Cell Spacing – Table Borders.	6

TEXTBOOKS:

1. Alex is Leon & Mathews Leon, “Internet for Everyone”, Leon Tech World, Chennai.
2. Eric Kramer, “HTML”.

REFERENCE BOOKS:

1. Kamalesh N.Agarwala, Amit Lal& Deeksha Agarwala,“Business of the net”.
2. JohnZabour,JeffFoust&DavidKerven,“HTML 4 HOW-TO”.

WEB RESOURCES:

- World Wide Web Consortium (W3C)
- Mozilla Developer Network (MDN Web Docs)
- W3Schools

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
CO1	Explain web-related concepts and architectures	**K1**
CO2	Describe Internet addressing and protocols	**K2**
CO3	Use web searching and e-mail management tools	**K3**
CO4	Design basic web pages using HTML	**K4**
CO5	Create HTML tables and understand XML fundamentals	**K5**

THIRD YEAR	CORE COURSE–V	COURSE CODE: 25CSCC5
SEMESTER V	MICROPROCESSOR AND MICROCONTROLLER	CREDITS: 5

COURSE OBJECTIVES:

- TointroducetheinternalorganizationofIntel8085Microprocessor.
- Toenablethestudentstowriteassemblylanguageprogramsusing8085.
- Tointerfacetheperipheraldevicesto8085usingInterrrupt controller and DMA interface.
- To provide real-life applications using microcontroller.

UNITS	CONTENTS	HOURS
I	Digital Computers - Microcomputer Organization-Computer languages –Microprocessor Architecture and its operations – Microprocessor initiated operations and 8085 Bus organization – Internal Data operations and 8085 registers - Peripheral or External initiated operations.	20
II	8085 Microprocessor – Pinout and Signals – Functional block diagram - 8085 Instruction Set and Classifications.	20
III	BCD to Binary and Binary to BCD conversions - ASCII to BCD and BCD to ASCII conversions- Binary to ASCII and ASCII to Binary conversions. BCD Arithmetic-BCD addition and Subtraction – Multi byte Addition and Subtraction - Multiplication and Division.	20
IV	The8085Interrupts–RIMANDSIMinstructions-8259 –Programmable Interrupt Controller- Direct Memory Access (DMA) and 8257 DMA controller.	15
V	Introduction to Microcontroller - Microcontroller Vs Microprocessor - 8051 Microcontroller architecture -8051 pin description. Timers and Counters – Operating Modes- Control Registers. Interrupts – Interrupts in 8051 - Interrupts Control Register – Execution of interrupt.	15

TEXT BOOKS:

1. R.S.Gaonkar-"Microprocessor Architecture- Programming and Applicationswith8085"-5thEdition-Penram International Publications,2021. [For unit I to unit IV].
2. SoumitraKumarMandal-"MicroprocessorsandMicrocontrollers–Architectures,Programmingand Interfacing using 8085, 8086, 8051", Tata McGraw Hill Education Private Limited. [for unit V].

REFERENCE BOOKS:

1. Mathur-"IntroductiontoMicroprocessor"-3rdEdition-TataMcGraw-Hill-2020.
2. RajKamal-"Microcontrollers:Architecture,Programming,InterfacingandSystemDesign",Pearson Education, 2025.
3. KrishnaKant,"MicroprocessorsandMicrocontrollers–Architectures,ProgrammingandSystem Design 8085, 8086, 8051, 8096", PHI, 2020.

WEBRESOURCES:

1. Web resources from NDL Library, E-content from open source libraries

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
CO1	RemembertheBasicbinarycodesandtheirconversions.Binaryconceptsareusedin Microprocessor programming and provide a good understanding of the architecture of 8085.	K1
CO2	Understanding the8085instructionset and their classifications, enables the students to write the programs easily on their own using different logic..	K2
CO3	Applying different types of instructions to convert binary codes and analyzing the outcome. The instruction set is applied to develop program son multiple by arithmetic operations.	K3
CO4	Analyze how peripheral devices are connected to 8085usingInterruptsandDMAcontroller.	K4
CO5	An exposure to create real time application susing microcontroller.	K5

THIRD YEAR	CORE COURSE–VI	COURSE CODE: 25CSCC6
SEMESTER V	OPERATING SYSTEMS	CREDITS: 5

COURSE OBJECTIVES:

- To provide the Fundamental Concepts in an Operating System.
- To study various memory management techniques including paging, segmentation, and virtual memory.
- To analyze process scheduling algorithms, synchronization mechanisms, deadlocks, and multiprocessing concepts.
- To understand device management, I/O systems, and storage device operations.
- To learn file management techniques, file organization, access methods, and protection mechanisms.

UNITS	CONTENTS	HOURS
I	Introducing Operating Systems Introduction - What Is an Operating System-Operating System Software -A Brief History of Machine Hardware -Types of Operating Systems -Brief History of Operating System Development-Object-Oriented Design	20
II	Memory Management Early Systems: Single-User Contiguous Scheme -Fixed Partitions-Dynamic Partitions Best-Fit versus First-Fit Allocation -Deallocation -Relocatable Dynamic Partitions. Virtual Memory: Paged Memory Allocation-Demand Paging-Page Replacement Policies and Concepts -Segmented Memory Allocation-Segmented/Demand Paged Memory Allocation - Virtual Memory-Cache Memory	20
III	Processor Management Overview-About Multi-Core Technologies-Job Scheduling Versus Process Scheduling Process Scheduler-Process Scheduling Policies-Process Scheduling Algorithms -A Word About Interrupts-Deadlock-Seven Cases of Deadlock -Conditions for Deadlock Modeling Deadlock-Strategies for Handling Deadlocks –Starvation Concurrent Processes: What Is Parallel Processing-Evolution of Multiprocessors Introduction to Multi-Core Processors-Typical Multiprocessing Configurations--Process Synchronization Software	20
IV	Device Management Types of Devices-Sequential Access Storage Media-Direct Access Storage Devices Magnetic Disk Drive Access Times- Components of the I/O Subsystem Communication among Devices-Management of I/O Requests	15
V	File Management The File Manager -Interacting with the File Manager -File Organization - Physical Storage Allocation -Access Methods-Levels in a File Management System - Access Control Verification Module	15

TEXT BOOKS:

1. Understanding Operating Systems, Ann McIver McHoes and Ida M. Flynn, Course Technology, Cengage Learning, 2021.

REFERENCE BOOKS:

1. OperatingSystems,AchyutGodbole and AtulKahate , McGraw Hill Publishing, 2022.
2. Operating Systems by Harvey M. Deitel, Paul J. Deitel, and David R. Choffnes, Pearson Education 2021.

WEBRESOURCES:

1.https://pdos.csail.mit.edu/6.1810/2024/xv6.html?utm_source

2.https://www.geeksforgeeks.org/operating-systems/?utm_source=chatgpt.com

COURSE OUTCOMES:

CO #	CO Statement	K-Level
CO1	Recall and describe the foundational history, architecture, types of operating systems, and object-oriented design concepts.	K1
CO2	Explain and contrast early contiguous memory schemes, dynamic partition layouts, and the mechanics of virtual memory management.	K2
CO3	Apply and calculate processor scheduling algorithms, page replacement policies, and dynamic memory allocation strategies.	K3
CO4	Analyze and model deadlock conditions, parallel multiprocessing configurations, and magnetic disk access constraints.	K4
CO5	Evaluate and justify file allocation methods, input/output device request pipelines, and access control security frameworks.	K5

THIRD YEAR	CORE COURSE VII	COURSECODE: 25CSCC7
SEMESTER V	WEB TECHNOLOGY	CREDITS: 5

COURSE OBJECTIVES:

- Design dynamic web pages using Java script, Jquery and Angular Java script
- Develop Web pages using HTML, CSS and XML
- Create web application using PHP and My SQL
- To design dynamic web pages using Angular java script
- Develop interactive web pages using Jquery

UNITS	CONTENTS	HOURS
I	INTRODUCTION TO XHTML AND CSS: Basic syntax, Standard structure, Basic text-markup, Images, Hypertext Links. Lists, Tables ,Forms, Frames, syntactic differences between HTML and XHTML-Introduction, Levels of style sheets, Style specification formats, Selector forms, Property value forms, Fontproperties,List properties,Color,Alignment of text,The box model, Background images.	15
II	THE BASICS OF JAVASCRIPT: Overview of JavaScript, Object orientation and JavaScript, general Syntactic characteristics, Primitives, operations, and expressions, Screen output and keyboard input, Controlstatements,Objectcreationandmodification,Arrays,Functions,Constructors,Patternmatchingusingregular expressions, Errors in scripts.	15
III	DYNAMIC DOCUMENTS WITH JAVA SCRIPT AND XML:Introduction,PositioningElements,Movi ng Elements, Element Visibility,Changing Color and Fonts, Dynamic Content,StackingElements,LocatingtheMouseCursor,ReactingtoaMouseClick,SlowMovementofElements. Introduction to XML, Syntax of XML, XML Document Structure, Document type definitions, Namespaces, XML schemas.	15
IV	PHP: Introduction to PHP: Overview of PHP- General Syntactic Characteristics- Primitives, Operations, and Expressions- Output-Control Statements- Arrays- Functions-Pattern Matching- Form Handling-Cookies-Session Tracking.	15
V	ANGULARJS AND JQUERY: Introduction JQuery, Syntax, selectors, events, JQuery HTML, JQuery Effects,JQuery CSS. Introduction to AngularJS, Directives, Expressions, Controllers, Filters, Services, Events, Forms, Validations, Examples.	15

TEXT BOOKS:

1. RobertW.Sebesta:ProgrammingtheWorldWideWebEighthEdition,Pearsoneducation,2021.UNITS:1,2,3,4
2. DayleyBrad,DayleyBrendan,"AngularJS,JavaScript,andjQueryAllinOne",SamsTeachYourself 1stEdition,KindleEdition, 2024. UNIT: 5

REFERENCE BOOKS:

1. M.Srinivasan: Web Programming Building Internet Applications, 3rdEdition,WileyIndia,2022.
2. Jeffrey C. Jackson: Web Technologies-A Computer Science Perspective, PearsonEducation,7thImpression,2022.
3. ChrisBates: Web Technology Theory and Practice ,Pearson Education,2021.
4. RajKamal: Internet and Web Technologies, McGraw Hill Education.

COURSE OUTCOMES

CO #	CO Statement	K-Level
CO1	Understand and apply the fundamentals of web page design using XHTML and CSS, including page structure, text formatting, hyperlinks, tables, forms, frames, and styling techniques.	K1
CO2	Develop responsive and visually appealing web pages using CSS concepts such as selectors, box model, fonts, colors, alignment, lists, and background images.	K2
CO3	Analyze and implement client-side scripting using JavaScript, including control structures, functions, arrays, objects, constructors, regular expressions, and error handling.	K3
CO4	Create dynamic and interactive web applications using JavaScript and XML techniques such as DOM manipulation, event handling, element positioning, animations, and XML document processing.	K4
CO5	Design server-side web applications using PHP by implementing form handling, session tracking, cookies, arrays, functions, pattern matching, and database-oriented scripting concepts.	K5

THIRD YEAR	CORE PRACTICAL- V	COURSE CODE: 25CSCP5
SEMESTER V	MICROPROCESSOR AND MICROCONTROLLER LAB	CREDITS:3

COURSE OBJECTIVES :

- To introduce the internal organization fIntel 8085 Microprocessor.
- To enable the students to write assembly language programs using 8085.
- To interface the peripheral devices to 8085 using Interrupt controller and DMA interface.
- To provide real –life applications using micro controller.

LIST OF PROGRAMS:

I. Addition and Subtraction

1. 8-bitaddition
2. 16-bitaddition
3. 8-bitsubtraction
4. BCD subtraction

II. Multiplication and Division

1. 8-bitmultiplication
2. BCD multiplication
3. 8-bitdivision

III. Sorting and Searching

1. Searching for an element in an array.
2. Sorting in Ascending and Descending order.
3. Finding the largest and smallest elements in an array.
4. Reversing array elements.

IV. Code Conversion

1. Binary to ASCII and ASCII to binary
2. ASCII to BCD and BCD to ASCII

V. Simple programs on 8051 Microcontroller

1. Addition
2. Subtraction
3. Multiplication
4. Division

COURSE OUTCOMES:

Remember the Basic binary codes and their conversions. Binary concepts are used in Microprocessor programming and provide a good understanding of the architecture of 8085.

CO #	CO Statement	K-Level
C01	Understanding the 8085-instruction set and their classifications,enables the students to write the programs easily on their own using different logic.	K1
C02	Applying different types of instruction to convert binary codes and analyzing the outcome.	K2
C03	The instruction set is applied to develop programs on multi byte arithmetic operations.	K3
C04	Analyze how peripheral devices are connected to 8085 using Interrupts and DMA controller.	K4
C05	An exposure to create real time applications using microcontroller.	K5

THIRD YEAR	MAJOR BASED ELECTIVE–I	COURSECODE: 25CSMB1A
SEMESTER -V	A.VIRTUALIZATION AND CLOUD	CREDITS: 4

COURSE OBJECTIVES:

- To understand basic concepts of distributed computing,
- To Provideagoodunderstanding oftheconcepts,standardsinCloudcomputing
- Tomakethestudentunderstandaboutthecloudserviceprovidersandtheirusage.
- Tolearnhowtosecurethe datainclouddepending.
- Tounderstandthevariousservicelevelagreements.

UnderstandthebasicprinciplesofCloudComputing,VirtualizationandDatacenters

UNITS	CONTENTS	HOURS
I	DISTRIBUTEDSYSTEMS Distributeasystem -Distributedalgorithm -DistributedDataStores - DistributedComputing-File Systems - Distributed Messaging - Distributed Applications – Distributed Transaction -Parallelanddistributedcomputing-Applications.	15
II	CLOUDCONCEPTS IntroductionCloudComputing- AdvantagesofCloud-PublicCloud-fiveessentialcharacteristics - three service models - Four deployment models - Benefits of Cloud Computing	15
III	VIRTUALIZATION IntroductiontoVSphereandtheSoftware - DefinedDataCenterCreatingVirtualMachines - VCenter Server - Configuring and Managing -Virtual Storage - Virtual Machine Management -.	15
IV	VIRTUALMACHINES VSphereHA-vSphereFault Tolerance-ProtectingDatavSphereDRS-NetworkScalability-Storage Scalability - Securing Virtual Machines.	15
V	DATACENTER Datacenteroverview-Components-DataCenter-Architecture-DifferentRacks-Datacenter Architecture for cloud computing - role of data center in cloud computing.	15

TEXTBOOKS:

1. JeanDollimoreformerlyofQueenMary,TimKindberg,"DistributedSystemsConcepts and Design", 5th Edition Cambridge University, University of London

REFERENCE BOOKS:

1. VenkataJosyula,MalcolmOrr,GregPage,"CloudComputing:Automatingthe Virtualized Data Center", 1st Edition.

2. BrianJ.S.Chee,CurtisFranklinJr.,"CloudComputing:TechnologiesandStrategiesof the Ubiquitous Data Center", 1st Edition.

WEB RESOURCES:

1. **https://www.upgrad.com/tutorials/software-engineering/software-key-tutorial/virtualization-in-cloud-computing/?utm_source**
2. **https://app.sophia.org/tutorials/virtualization-and-cloud-os?utm_source=chatgpt.com**

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
C01	Understand what is Cloud Computing.	K1
C02	Know the architecture of the cloud and the usage of clouds.	K3
C03	Secure their data from the security issues.	K3
C04	Understand What is Virtualization.	K4
C05	Understand Cloud Types and Cloud Service Deployment Models (IaaS*, PaaS*, SaaS*).	K5

THIRD YEAR	MAJOR BASED ELECTIVE–I	COURSE CODE: 25CSMB1B
SEMESTER V	B.BIG DATA ANALYTICS	CREDITS: 4

COURSE OBJECTIVES:

- To know the fundamental concepts of big data and analytics.
- To understand the evolution, characteristics, architecture, and applications of Big Data systems.
- To learn distributed storage and processing frameworks such as HDFS, Map Reduce, and YARN.
- To apply clustering and classification techniques for analyzing large-scale datasets.
- To understand association rule mining and recommendation system methodologies.

UNITS	CONTENTS	HOURS
I	**INTRODUCTION TO BIG DATA** : Evolution of Big data — Best Practices for Big data Analytics — Big data characteristics — Validating — The Promotion of the Value of Big Data — Big Data Use Cases- Characteristics of Big Data Applications — Perception and Quantification of Value - Understanding Big Data Storage — A General Overview of High-Performance Architecture — HDFS — Map Reduce and YARN — Map Reduce Programming Model	15
II	**CLUSTERING AND CLASSIFICATION :**Advanced Analytical Theory and Methods: Overview of Clustering — K-means — Use Cases — Overview of the Method — Determining the Number of Clusters — Diagnostics — Reasons to Choose and Cautions .- Classification: Decision Trees — Overview of a Decision Tree — The General Algorithm — Decision Tree Algorithms — Evaluating a Decision Tree — Decision Trees in R — Naïve Bayes — Bayes? Theorem — Naïve Bayes Classifier	15
III	**ASSOCIATION AND RECOMMENDATION SYSTEM:** Advanced Analytical Theory and Methods: Association Rules — Overview — Apriori Algorithm — Evaluation of Candidate Rules — Applications of Association Rules — Finding Association& finding similarity — Recommendation System: Collaborative Recommendation- Content Based Recommendation — Knowledge Based Recommendation- Hybrid Recommendation Approaches	15
IV	**STREAM MEMORY:** Introduction to Streams Concepts — Stream Data Model and Architecture — Stream Computing, Sampling Data in a Stream — Filtering Streams — Counting Distinct Elements in a Stream — Estimating moments — Counting oneness in a Window — Decaying Window — Real time Analytics Platform(RTAP) applications —	15

	Case Studies — Real Time Sentiment Analysis, Stock Market Predictions. Using Graph Analytics for Big Data: Graph Analytics.	
V	**NO SQL DATA MANAGEMENT FOR BIG DATA AND VISUALIZATION** : NoSQL Databases :Schema-less Models?: Increasing Flexibility for Data Manipulation-Key Value Stores- Document Stores - Tabular Stores - Object Data Stores -Graph Databases Hive -Sharding -Hbase - Analyzing big data with twitter -Big data for E-Commerce Big data for blogs - Review of Basic Data Analytic Methods using R.	15

TEXTBOOKS:

1. AnandRajaramanandJeffreyDavidUllman,"MiningofMassiveDatasets",Cambridge University Press, 2021.
2. NoSQL Distilled ,Addison-Wesley Professional.,addision publisher 2022.

REFERENCE BOOKS:

1.DavidLoshin,"BigDataAnalytics:FromStrategicPlanningtoEnterpriseIntegrationwith Tools, Techniques, NoSQL, and Graph", Morgan Kaufmann/El sevier Publishers, 2022.

2.EMCEducationServices,"DataScienceandBigDataAnalytics:Discovering,Analyzing, Visualizing and Presenting Data", Wiley publishers, 2025.

WEB RESOURCES:

1. WebresourcesfromNDLLibrary,E-contentfromopen-source libraries.
2. https://hadoop.apache.org/docs/?utm_source
3. https://www.kaggle.com/learn?utm_source

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
C01	Understand Big Data technologies and architectures.	K1
C02	Apply machine learning and data mining techniques.	K3
C03	Process and analyze streaming and large-scale datasets.	K3
C04	Use Hadoop, NoSQL, Hive, and HBase tools effectively.	K4
C05	Develop Big Data solutions for real-world applications.	K5

THIRD YEAR	SKILL BASEDELECTIVE–I	COURSE CODE: 25CSSB1A
SEMESTER V	A.AMAZON WEB SERVICES	CREDITS: 2

COURSE OBJECTIVES:

- To understand the fundamentals, history, uses, and advantages of Amazon Web Services (AWS).
- To learn cloud infrastructure services such as EC2, EBS, Load Balancing, and AMI management.
- To understand AWS storage services including S3, lifecycle management, CloudFront CDN, and Storage Gateway.
- To study AWS messaging and application integration services such as SQS, SNS, SWF, API Gateway, and Kinesis.
- To understand AWS database services including RDS, DynamoDB, Aurora, and Redshift.

UNITS	CONTENTS	HOURS
I	What is AWS?-uses-Advantages of AWS-History- AWS free tier.	6
II	AWS EC2-AWS EBS-AWS Load Balancing-creating Load Balance-AWS AMI.	6
III	AWS S3-AWS S3 concept-creating S3 Bucket-life cycle management-cloud front CDN-storage gateway.	6
IV	AWS SQS-AWS SWF-AWS SNS-API Gateway-AWS Kinesis.	6
V	AWS database services-relational database-nonrelational database-creating an RDS instance-Dynamo DB-AWS Aurora-AWS red shift.	6

TEXTBOOKS:

1. AmazonWebServices,firstedition,K.Strong,Volume-II

REFERENCE BOOKS:

1.AmazonWebServiceshandbookBasicamazonwebservicesreferenceforcloudarchitect, Jarrett cross.

WEB RESOURCES:

https://www.javatpoint.com/aws-tutorial

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
C01	Explain the concepts, benefits, history, and applications of Amazon Web Services (AWS).	K1
C02	Understand and configure AWS compute services such as EC2, EBS, Load Balancer, and Amazon Machine Images (AMI).	K3
C03	Apply AWS storage solutions including S3 buckets, lifecycle management, CloudFront CDN, and Storage Gateway.	K3
C04	Analyze AWS messaging, workflow, and streaming services such as SQS, SWF, SNS, API Gateway, and Kinesis.	K4
C05	Understand relational and non-relational AWS database services including RDS, DynamoDB, Aurora, and Redshift.	K5

THIRD YEAR	SKILL BASED ELECTIVE – I	COURSE CODE: 25CSSB1B
SEMESTER V	B.WEB DESIGNING	CREDITS: 2

COURSE OBJECTIVES:

- To understand the basic concepts of computer networks, client–server architecture, and web technologies.
- To learn Internet fundamentals including IP addressing, domain names, and communication protocols (HTTP, FTP, SMTP, etc.).
- To gain knowledge of web browsing, search engines, email systems, and web navigation techniques.
- To design web pages using HTML including text formatting, lists, images, links, and tables.
- To understand XML fundamentals and differentiate between HTML and XML.

UNITS	CONTENTS	HOURS
I	Introduction to Web related terms: Client –Server–Single Tier Architecture–Two Tier Architecture– N Tier Architecture – Internet – Intranet – WWW – Web Page – Web Site.	6
II	Introduction to Internet and its Protocols: Internet Addressing – IP Address – Domain Name – Electronic Mail–Uniform Resource Locator –TCP/IP-FTP–HTTP–SHTTP-SMTP–POP–UDP – PPP – Telnet – Gopher – WAIS.	6
III	SearchingtheWeb:Browser:WebIndex–WebSearchEngine–MetaSearchSites–Directoryand Indexes – Managing Mails: Address Book – File Attachment Facility – Junk Mail.	6
IV	Introductionto HTML: HTMLCodeforWebPage–HTMLTags-WebPageBasics–DisplayaWeb Page in a Web Browser – Format texts – Font Style and sizes – Lists – Ordered Lists – Unordered Lists – Image – Add an image – Background Image – Horizontal Rule.	6
V	Introduction to HTML Table: table- CellPadding – Cell Spacing – TableBorders. IntroductiontoXML:CreatingBasicXMLProgram–XMLVsHTML–Advantagesand Disadvantages of XML – XML Parser- Introduction to CSS.	6

TEXTBOOKS:

1. AlexisLeon&MathewsLeon, "InternetforEveryone", LeonTechWorld,Chennai.2.
2. EricKramer,"HTML".KamaleshN.Agarwala,AmitLal&DeekshaAgarwala,"Businessofthenet".
3. JohnZabour,JeffFoust&DavidKerven,"HTML4HOW-TO".

WEB RESOURCES:

1. https://www.w3schools.com/html/?utm_source
2. https://developer.mozilla.org/en-US/docs/Web/HTML?utm_source
3. https://learn.microsoft.com/en-us/exchange/mail-flow/mail-flow?utm_source

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
C01	Explain the fundamentals of Internet, WWW, web architecture, and networking concepts.	**K1**
C02	Identify and describe various Internet protocols and their roles in web communication.	**K3**
C03	Use web browsers, search engines, and email systems effectively for information retrieval and communication.	**K3**
C04	Design basic web pages using HTML elements such as text formatting, lists, images, and links.	**K4**
C05	Create and manage HTML tables and apply formatting techniques for structured data representation.	**K5**

THIRD YEAR	INTERNSHIP	COURSE CODE: 25CSIS
SEMESTER- V		CREDITS: 2

COURSE OBJECTIVES:

- Provide practical exposure to real-world working environments and industry practices.
- Apply theoretical knowledge to solve real-time problems in the chosen domain.
- Develop technical, analytical, and problem-solving skills through hands-on experience.
- Enhance professional communication, teamwork, and workplace ethics.
- Gain experience in project execution, documentation, and presentation of outcomes.

S.NO	INTERNSHIP EVALUATION
I	Report of the Study • Introduction to organization and domain • Objectives and scope of the internship • Description of work carried out / project details • Tools, technologies, and methodologies used • Results, analysis, and outcomes • Challenges faced and solutions implemented • Learning outcomes and skills acquired • Report presentation (format, clarity, documentation quality)
II	Viva Voce • Understanding of the work/project done • Conceptual clarity and technical knowledge • Ability to explain tools and technologies used • Problem-solving approach during internship • Communication and presentation skills

COURSE OUTCOMES:

CO #	CO Statement	K-Level
CO1	Understand real-world organizational structure and professional work environment.	**K2**
CO2	Apply theoretical knowledge to solve practical problems in the workplace.	**K3**
CO3	Analyze tasks, challenges, and workflows in the chosen domain.	**K4**
CO4	Develop and present a structured internship report with technical clarity.	**K6**
CO5	Evaluate personal performance, skills gained, and professional growth during the internship.	**K5**

Upon successful completion of this course the students would be able to:

THIRD YEAR	CORE COURSE–VIII	COURSE CODE: 25CSCC8
SEMESTER VI	COMPUTER NETWORKS	CREDITS: 5

COURSE OBJECTIVES:

- To describe the general principles of Computer Networks.
- To describe how the different layers in a computer network work
- To know about Wired LAN: IEEE Standards and Satellite networks.
- To understand data link layer protocols including error detection, flow control, and framing.
- To study LAN, WAN, wireless networks, and network interconnection devices.

UNITS	CONTENTS	HOURS
I	Data Communication – Networks – The Internet – Protocols and Standards – OSI model- Layers in OSI Model - TCP/IP Protocol Suite – Addressing.	20
II	Analog and Digital – Digital Signals – Transmission Impairment – Performance – Multiplexing – Guided Media – Unguided Media. Switching: Circuit Switched Networks –Datagram Networks – Virtual Circuit Networks	20
III	Data Link Layer: Error Detection and Correction -Introduction – Block Coding: Error detection, Error correction – Data Link Control: Framing – Flow and Error Control – Protocols – Noiseless Channels – Noisy channels – HDLC – Point to Point Protocol.	20
IV	Wired LAN: IEEE Standards – Standard Ethernet. Wireless LAN: IEEE 802.11 – Bluetooth. Connecting LANs: Connecting Devices – Virtual LANs. Wireless WAN: Cellular Telephony – Satellite Networks. Network Layer-Logical Addressing: IPv4 Addresses – IPv6 Addresses.	15
V	Transport Layer: Process to Process Delivery – User Datagram Protocol - TCP. Application Layer: Domain Name Space – DNS in the Internet – Electronic Mail – File Transfer. WWW: Architecture – HTTP.	15

TEXTBOOKS:

1. "Data and Computer Communications" by William Stallings A comprehensive book covering all fundamental concepts of data communication, networking protocols, and standards.
2. "Computer Networks" by Andrew S. Tanenbaum and David J. Wetherall Known for clear explanations and in-depth coverage of networking principles, protocols, and architectures like OSI and TCP/IP.
3. "Data Communications and Networking" by Behrouz A. Forouzan Suitable for beginners and intermediate learners, covering essential topics with detailed explanations and diagrams.

REFERENCE BOOKS:

1. Behrouz A. Forouzan, “Data Communications and Networking”, McGraw-Hill Companies, New York, 8th Edition, 2022.
2. William Stallings “Data and computer communications”, Prentice Hall of India, 8th Edition, 2024.
3. Andrew S Tanenbaum, “Computer Networks”, Prentice Hall of India, New Delhi,2024.

WEB RESOURCES:

1.https://www.guru99.com/data-communication-computer-network-tutorial.html

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
C01	Explain basic concepts of data communication, networking models, and Internet architecture.	K1
C02	Analyze OSI and TCP/IP models and understand protocol layering and addressing mechanisms.	K3
C03	Evaluate transmission media, signal types, and performance issues in data communication systems.	K3
C04	Identify the appropriate protocols and standards for computer networks.	K4
C05	Classify technical factors of cellular networks and satellite communication.	K5

THIRD YEAR	CORE COURSE–IX	COURSE CODE: 25CSCC9
SEMESTER VI	MOBILE APPLICATION DEVELOPMENT	CREDITS: 5

COURSE OBJECTIVES:

- To learn the basic concepts of mobile app using Android Platform
- To learn the detail framework of Android Platform
- To learn the design aspects in Creating intuitive, reliable mobile apps using the android services and components
- To understand the various additional features of mobile app development
- To learn Graphical features to design attractive web apps

UNITS	CONTENTS	HOURS
I	**Introduction to Android Platform** Introducing the Android Software Development Platform: Understanding Java SE and the Dalvik Virtual Machine-The Directory Structure of an Android Project-Common Default Resources Folders-The Values Folder- Leveraging Android XML- Screen Sizes-Desktop Clocks- Using Android Application Resources-Launching Application: The Android Manifest.xml File - Creating Your First Android Application-Running the App-Adding an Application Icon-Adding Transparency.	20
II	**Overview of Android Framework** Android Framework Overview: The Foundation of OOP: The Object-The Blue print for an Object: The Class-Providing Structure for Classes: Inheritance-Defining an Interface Bundling Classes-An overview of XML- The APK File-Android Application Components Android Activities- Android Services - Broad cast Receivers - Content Providers - Android Manifest XML.	20
III	**Working with Screen Layouts** Screen Layout Design- Android View Hierarchies - Nesting Views-Defining Screen Layouts - Editing the main.xml File-Using Relative Layouts- Sliding Drawers-Using Padding and Margins with Views and Layouts.	20
IV	**Working with Buttons, Menus, and Dialogs** UI Design: Buttons, Menus, and Dialogs: Using Common UI Elements- Adding an Image Button to Your Layout-Defining Multistate Image Button - Graphics in XML -Editing the main.xml File-Replacing the Default Background- Adding a Text to Your Layout-Adding an Image-Using Menus in Android-Creating the Menu Structure with XML-Running the Application in the Android Emulator- Making the Menu Work-Adding Dialogs.	15
V	**Introduction to Graphics in Android** An Introduction to Graphics Resources in Android: Introducing the Drawables-Implementing Images - Creating Animation in Android- Tween Animation in Android-Using Transitions - Creating 9-Patch Custom Scalable Images-Playing Video in Android Apps- SQLite based simple applications	15

TEXTBOOKS:

1. "Android Cookbook: Problems and Solutions for Android Developers" Ian Darwin, Mark L. Murphy

2. "Android Graphics, Animation and UI" (par exemple, "Android Graphics, Animation and UI" de Wallace Jackson)

REFERENCE BOOKS:

1. Smith, D., & Friesen, J. (2021). Android Recipes: A Problem -Solution Approach, (1st Ed.). Rakmo Press.
2. DiMarzio, J. F. (2022). Android: A Programmer's Guide, (1st Ed.). Tata Mcgraw Hill.
3. Murphy, M. L. (2022). The Busy Coder's Guide to Android Development, (3rd ed.). Commons Ware.

WEB RESOURCES:

1.Android Developer's Guide - available at: http://developer.android.com/

COURSE OUTCOMES:

CO #	CO Statement	K-Level
CO1	Recall the elements of software development platform to build android programming.	**K1**
CO2	understand the UI-component layouts, event handling, and screen orientations to develop mobile applications.	**K2**
CO3	Apply the Screen Layout and UI Design in Android Framework to implement the android apps.	**K3**
CO4	organize the various resources and examine the parameter passing mechanism among them.	**K4**
CO5	design and evaluate the user interfaces to support mobile application development.	**K5**

THIRD YEAR	CORE COURSE–IV	COURSE CODE: 25CSCC9
SEMESTER VI	MOBILE APPLICATION DEVELOPMENT LAB	CREDITS: 4

COURSE OBJECTIVES:

1. To understand the fundamentals of Android application development and its architecture.
2. To design user interfaces using various layouts such as Linear, Relative, and Constraint layouts.
3. To implement event-driven programming concepts in Android applications.
4. To develop applications using intents for communication between activities.
5. To work with Android components such as fragments, media players, and animations.

LIST OF PROGRAMS:

1. Different Layout design including nested layout
2. Arithmetic Operations
3. Business Calculator
4. Animation: Bouncing of a ball
5. Intent Creation.
6. Prepare Student Bio-data using Database SQLite
7. Fragments-Tablet Programming
8. Media Player

COURSE OUTCOMES:

CO #	CO Statement	K-Level
CO1	Understand the Android platform architecture, development environment, and application structure.	K1
CO2	Design and develop Android user interfaces using different layouts and UI components.	K2
CO3	Implement arithmetic and business logic applications using event handling techniques.	K3
CO4	Apply intents to enable communication between multiple activities in Android applications	K4
CO5	Design fragment-based applications suitable for tablets and large-screen devices.	K5

THIRD YEAR	MAJOR BASED ELECTIVE-II	COURSE CODE: 25CSMB2A
SEMESTER VI	A. CLOUD COMPUTING	CREDITS: 4

COURSE OBJECTIVES:

- To impart fundamental concepts of Cloud Computing.
- To impart a working knowledge of the various cloud service types and their uses and pit falls.
- To enable the students to know the common features and differences in the service offerings of the three major Cloud Computing service providers, namely Amazon, Microsoft and Google.
- To provide know-how of the various aspects of application design, bench marking and security on the Cloud.

UNITS	CONTENTS	HOURS
I	Cloud computing Foundation: Introduction to Cloud Computing–Move to Cloud Computing–Types of Cloud–Working of Cloud Computing.	15
II	Cloud Computing Architecture: Cloud Computing Technology – Cloud Architecture–Cloud Modeling and Design- Virtualization: Foundation–Grid,Cloud and Virtualization–Virtualization and CloudComputing	15
III	DataStorage and Cloud Computing: DataStorage–CloudStorage–CloudStorage from LANs to WANs – Cloud Computing Services : Cloud Services – CloudComputingatWork	15
IV	Cloud Computing and Security : Risks in Cloud Computing – Data Security inCloud–Cloud SecurityServices–Cloud Computing Tools: Tools and Technologies for Cloud–Cloud Mashaps–Apache Hadoop–CloudTools.	15
V	Cloud Applications–Moving Applications to the Cloud – Microsoft CloudServices– Google Cloud Applications–Amazon CloudServices– Cloud Applications	15

TEXT BOOKS:

1.CloudComputing–A Practical Approach for Learning andImplementation, A.Srinivasan and J.Suresh,Pearson India Publications,2024

REFERENCE BOOKS:

1.Cloud computing :Principles and Paradigms edited by Rajkumar Buyya,James Broberg, Andrezej, Wiley india publications 2021

WEB RESOURCES:

COURSE OUTCOMES:

CO #	CO Statement	K-Level
CO1	Understand the fundamentals of cloud computing, including evolution, types of clouds, and working principles.	**K1**
CO2	Analyze cloud computing architecture, cloud design models, and deployment strategies.	**K2**
CO3	Explain virtualization concepts and their relationship with grid and cloud computing environments.	**K3**
CO4	Understand cloud storage systems and evaluate data storage mechanisms across LAN, WAN, and cloud environments.	**K4**
CO5	Identify and analyze cloud computing services and their applications in real-world environments.	**K5**

THIRD YEAR	MAJOR BASED ELECTIVE-II	COURSE CODE: 25CSMB2B
SEMESTER VI	B. INTRODUCTION OF DATA SCIENCE	CREDITS: 4

COURSE OBJECTIVES:

- To introduce the concepts, techniques and tools in Data Science.
- To understand the various facets of data science practice including data collection and integration, exploratory data analysis, predictive modeling, descriptive modeling and effective communication.
- To understand Hadoop, Spark, NoSQL databases, and distributed data processing frameworks.
- To analyze real-world case studies and apply Data Science techniques for prediction and decision-making.
- To develop practical skills in data preparation, analysis, visualization, and automation.

UNITS	CONTENTS	HOURS
I	**Introduction:**Benefitsanduses–Facetsofdata–Datascienceprocess–Bigdataecosystem and data science.	15
II	**The Data science process**: Overview–research goals- retrieving data-transformation–Exploratory Data Analysis – Model building.	15
III	**Algorithms**: Machine learning algorithms–Modeling process–Types–Supervised–Unsupervised - Semi-supervised.	15
IV	**Introduction to Hadoop**: Hadoop framework–Spark–replacing Map Reduce–NoSQL–ACID– CAP – BASE – types.	15
V	**Case Study**: Prediction of Disease- Setting research goals- Data retrieval–preparation- exploration- Disease profiling - presentation and automation.	15

TEXT BOOKS:

1.Davy Cielen,ArnoD.B. Meysman,MohamedAli,"Introducing Data Science",Manning publications 2021.

REFERENCE BOOKS:

1. RogerPeng,"TheArtofDataScience",lulu.com2021.
2. MurtazaHaider,"Getting StartedwithDataScience–MakingSenseofData with Analytics", IBM press, E-book.
3. Annalyn Ng,Kenneth Soo,"Numsense! Data Scienceforthe Layman: No Math Added", 2022,1st Edition.
4. CathyO'Neil,RachelSchutt,"DoingDataScienceStraightTalkfromtheFrontline", O'Reilly Media 2020.

WEB RESOURCES:

1. https://www.coursera.org/professional-certificates/google-data-analytics
2. https://www.data8.org/
3. https://seeing-theory.brown.edu/

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
CO1	Explain the concepts, benefits, and applications of Data Science and Big Data ecosystems.	**K1**
CO2	Apply the Data Science process including data retrieval, transformation, exploratory data analysis, and model building.	**K2**
CO3	Analyze and implement machine learning algorithms using supervised, unsupervised, and semi-supervised approaches.	**K3**
CO4	Understand Hadoop, Spark, NoSQL databases, and distributed computing technologies used in large-scale data processing.	**K4**
CO5	Compare ACID, CAP, and BASE properties in modern database systems and NoSQL environments.	**K5**

THIRD YEAR	SKILL BASED ELECTIVE II	COURSE CODE: 25CSSB2A
SEMESTER VI	A.CYBER SECURITY	CREDITS: 2

COURSE OBJECTIVES:

- ThestudentswillbeabletoUnderstandvariousblockcipherandstreamcipher models. Describe the principles of public key cryptosystems, hash functions and digital signature
- To get affirm knowledge on Cyber Security Essentials

UNITS	CONTENTS	HOURS
I	**Introduction to Security:** Data Encryption Standard-Block cipher principles-block cipher modes of operation-Advanced Encryption Standard (AES)-Triple DES-Blowfish-RC5 algorithm.	6
II	**Public Key Cryptography and Hash Algorithms:**Principlesofpublickeycryptosystems-TheRSAalgorithm-Keymanagement-DiffieHellmanKey exchange- Hash functions-Hash Algorithms (MD5, Secure Hash Algorithm	6
III	**Fundamentals of Cyber Security :**HowHackersCoverTheirTracks-FraudTechniques-ThreatInfrastructure-TechniquestoGainaFoothold (Shellcode, SQL Injection, Malicious PDF Files)- Misdirection, Reconnaissance, and Disruption Methods.	6
IV	**Planning for Cyber Security :** Privacy Concepts-Privacy Principles and Policies – Authentication and Privacy- Data Mining- Privacy on the Web - Email Security - Privacy Impacts of Emerging Technologies.	6
V	**Cyber Security Management :**Security Planning - Business Continuity Planning - Handling Incidents - Risk Analysis - Dealing with Disaster –Legal Issues – Protecting programs and Data –Information and the law – Rights of Employees and Employers - Emerging Technologies - The Internet of Things - Cyber Warfare.	6

TEXT BOOKS:

1.William Stallings,"CryptographyandNetworkSecurity",PearsonEducation,6thEdition,2021.

2.CharlesP. Pfleeger Shari LawrencePfleeger Margulies,SecurityinComputing,5th Edition, Pearson Education , 2024.

REFERENCE BOOKS:

1. Graham,J.Howard,R., Olson,R.,Cyber Security Essentials, CRCPress, 2021.
2. GeorgeK. Kostopoulous,CyberSpaceandCyberSecurity,CRCPress,2023.

WEB RESOURCES:

1. NPDLLibrary,E-contentfromopen-source libraries

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
CO1	Understand the fundamental concepts of cryptography, encryption techniques, and block cipher algorithms used in cyber security.	**K1**
CO2	Analyze symmetric and asymmetric cryptographic algorithms including DES, AES, Triple DES, Blowfish, RC5, and RSA.	**K2**
CO3	Apply public key cryptography, key exchange mechanisms, and hash algorithms such as MD5 and SHA for secure communication.	**K3**
CO4	Identify cyber threats, fraud techniques, hacker methodologies, and attack mechanisms including SQL injection and malicious code techniques.	**K4**
CO5	Understand privacy principles, authentication methods, email security, and web privacy issues in cyber environments.	**K5**

THIRD YEAR	SKILL BASED ELECTIVE II	COURSE CODE: 25CSSB2B
SEMESTER VI	B.ENTERPRISE RESOURCE PLANNING	CREDITS: 2

COURSE OBJECTIVES:

- To understand the concepts, evolution, structure, and importance of Enterprise Resource Planning (ERP) systems.
- To study ERP-related technologies such as Business Process Re-engineering, Data Warehousing, Data Mining, OLAP, and Supply Chain Management.
- To understand ERP applications in manufacturing, production planning, and materials management.
- To analyze the benefits and organizational impact of ERP implementation.
- To study ERP implementation lifecycle, project planning, testing, training, and maintenance processes.

UNITS	CONTENTS	HOURS
I	Introduction to ERP – Conceptual model of ERP – Evolution of ERP –Structure of ERP - Reasons for Growth – Advantages of ERP – Enterprise: An overview. ERP and Related Technologies: Business Process Re-engineering –Management Information System – Decision Support System – Executive Information system – Data Warehousing – Data Mining – OLAP – Supply Chain Management.	6
II	ERP – Manufacturing Perspective – Materials Requirement Planning –Distribution Requirements Planning – JIT – CAD/CAM – Product Data Management – ERP Modules: - Plant Maintenance – Quality Management –Materials Management.	6
III	Benefits of ERP: Reduction of Lead Time – Reduction of Cycle Time –Improved Resource Utilization – Reduced Quality Costs – Increased Flexibility –Improved Information accuracy and Decision making capability.	6
IV	ERP Implementation Lifecycle: Introduction – Per-evaluation screening –Project Planning – Gap Analysis – Reengineering – Configuration –Implementation – Testing – Training – Maintenance. Vendors, Consultants and Users: Inhouse Implementation-Pros and Cons – Vendors – Consultants – Endusers.	6
V	Future Directions in ERP: New Markets – New Channels – Faster implementation Methodologies – Business Models – Application Platforms – new business segments – Web-enabling – Market Snapshot.	6

TEXT BOOKS:

1.Alexis Leon , "Enterprise Resource Planning ",Tata McGraw – Hill Publishing Company Ltd.
2. Rahul V. Altekar , "Enterprisewide Resource planning-Theory and practice", Prentice Hall of India Pvt Ltd.
3. Vinod kumar garg and N.K.Venkitakrishnan , " Enterprisewide Resource prentice Hall of India Pvt Ltd.

WEB RESOURCES:

- **https://learning.sap.com/?utm_source**
- **https://www.netsuite.com/portal/resource/articles/erp/what-is-cloud-erp.shtml?utm_source**
- **https://www.forbes.com/sites/forbestechcouncil/2023/03/06/the-future-of-erp/?utm_source**

COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

CO #	CO Statement	K-Level
CO1	Explain the concepts, architecture, evolution, and significance of ERP systems in enterprises.	K1
CO2	Analyze ERP-related technologies including BPR, MIS, DSS, EIS, Data Warehousing, Data Mining, OLAP, and SCM.	K2
CO3	Understand ERP modules and manufacturing concepts such as MRP, DRP, JIT, CAD/CAM, and Product Data Management.	K3
CO4	Evaluate the organizational benefits of ERP systems including improved resource utilization, reduced cycle time, and enhanced decision-making.	K4
CO5	Describe the phases involved in ERP implementation lifecycle including planning, gap analysis, configuration, testing, training, and maintenance.	K5

THIRD YEAR	GENDER STUDIES	COURSE CODE: 25GS
SEMESTER VI		CREDIT: 1

UNITS	CONTENTS
I	INTRODUCTION TO GENDER STUDIES CONCEPTS Gender Spectrum.-Sex–Gender distinction–Biological Determinism–Patriarchy–Feminism–Gender Socialization and Stereotyping-Gender Discrimination–Gender Division of labour and roles–Gender Sensitivity and awareness–Gender Equity–Equality–Gender Main streaming and Gender Analysis.
II	UGC INITIATIVES ON WOMEN'S STUDIES Definition of Women's Studies–Gender Studies–UGC Initiatives and guidelines on Women's Studies - Beijing Conference, UN Initiatives–Convention on Elimination of All forms of Discrimination Against Women (CEDAW)- Sustainable Development Goals on Gender Equality (SDG 5) and targets
III	AREAS OF GENDER DISCRIMINATION Gender Socialization- Sex Ratio–Health and Nutrition––Literacy and Education - Employment- Governance–participation in decision making- politics- property rights and access to credit- gender based violence- Social institutions–Family, Caste, Class, religion, gender, State. Market–Media–Politics–Judiciary
IV	WOMEN DEVELOPMENT AND GENDER EMPOWERMENT Towards Equality Report of Status of Women in India 1974–International Women's Decade–International Women's Year–National Policy for Empowerment of Women 2001
V	WOMEN'S MOVEMENTS AND SAFEGUARDING MECHANISM & CURRENT CONTOURS: (for continuous internal assessment only) In India National /State Commission for Women(NCW) – All Women Police Station – Family Court Legislations safeguarding women –Transgender Policy—Constitutional amendments for women's political participation—Tamil Nadu State Policy for Women 2021- National Policy for Women 2015 – Prevention of Sexual Harassment at Work places Act 2013- Protection of Children from Sexual Offences Act, 2012 - Analysis of regressive and progressive High court and supreme court judgments- women proactive policies, programmes, interventions

THIRD YEAR	MINI PROJECT	COURSE CODE: 25CSPW
SEMESTER VI		CREDIT: 5

Students to do Mini Project in the irrespective Colleges .The **objective** of the Mini Project is to enable the students to work in convenient groups on to more than Four members on a project with a Latest Software.

COURSEOUTCOMES:

1. Students will be able to practice acquired knowledge within the chosen area of technology for project development.
2. Identify, discuss and justify the technical aspects of the chosen project with a comprehensive and systematic approach.

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20/05/2026

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